

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

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Biology

- 1 A metabolic enzyme is located exclusively inside the cytoplasm of living cells. When this enzyme comes into direct contact with a colourless water-soluble substrate, it catalyses a reaction that produces a blue product.

Assume that neither the enzyme nor the substrate can cross an intact cell surface membrane.

Four separate suspensions were prepared by adding the substrate to:

- mammalian cells in an isotonic solution
- mammalian cells in pure water
- intact plant cells in pure water
- plant protoplasts (plant cells from which cell walls have been enzymatically removed) in pure water

All four mixtures were incubated at 25 °C for 10 minutes.

Which row in the table correctly predicts whether a blue colour will develop in each mixture?

	mammalian cells in isotonic solution	mammalian cells in pure water	intact plant cells in pure water	plant protoplasts in pure water
A	blue colour develops	no colour change	no colour change	blue colour develops
B	blue colour develops	blue colour develops	no colour change	no colour change
C	no colour change	blue colour develops	no colour change	blue colour develops
D	no colour change	blue colour develops	blue colour develops	no colour change
E	no colour change	no colour change	blue colour develops	blue colour develops
F	no colour change	blue colour develops	no colour change	no colour change
G	blue colour develops	no colour change	blue colour develops	no colour change
H	no colour change	no colour change	no colour change	blue colour develops

- 2** The diagram shows a simplified transverse section through a dicotyledonous plant leaf, highlighting two distinct cellular layers, P and Q.

Which of the following statements is/are correct?

- 1 Layer P and Layer Q are both tissues.
- 2 The leaf is classified as an organ system.
- 3 Mitochondria are present in the cytoplasm of cells in both Layer P and Layer Q.

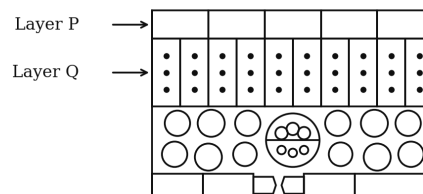


Diagram not to scale

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

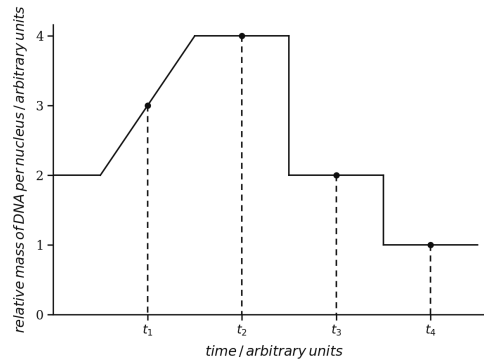
- 3** Which of the statements about the production of an insect-resistant transgenic crop plant are correct?
- 1 A restriction enzyme is used to cut the desired gene from the donor organism's DNA.
 - 2 DNA ligase is used to seal the gene into a plasmid vector.
 - 3 The genetically modified plant cells divide by meiosis during micropropagation to form new plantlets.
 - 4 All somatic cells in the mature regenerated plant will contain a copy of the inserted gene.
- A** 1 and 2 only
 - B** 1 and 4 only
 - C** 2 and 3 only
 - D** 3 and 4 only
 - E** 1, 2 and 4 only
 - F** 1, 3 and 4 only
 - G** 2, 3 and 4 only
 - H** 1, 2, 3 and 4

- 4 The graph shows the relative mass of DNA per nucleus during meiosis and gamete formation in a diploid animal.

Four time points are labelled t_1 , t_2 , t_3 , and t_4 .

Which of the following statements is/are correct?

- 1 Semi-conservative replication of DNA occurs during the period containing time t_1 .
- 2 At time t_3 , the nucleus contains pairs of homologous chromosomes.
- 3 The number of chromosomes in a nucleus at time t_4 is equal to the number of chromosomes in a nucleus at time t_3 .



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 5** β -thalassaemia is an inherited condition in humans caused by a mutation in the gene coding for the β -globin polypeptide of haemoglobin, leading to a deficiency of functional haemoglobin.

In one form of gene therapy, haematopoietic stem cells are removed from the bone marrow of a patient with β -thalassaemia. A functional β -globin gene is inserted into the nuclear DNA of these stem cells. The genetically modified stem cells are then returned to the patient, where they establish in the bone marrow and can differentiate into red blood cells.

Which of the following statements is/are correct?

- 1 The modified stem cells can divide by mitosis to produce replacement stem cells that also contain the inserted gene.
- 2 Mature red blood cells circulating in the patient's blood will each contain a nucleus carrying the inserted gene.
- 3 In developing red blood cells, mRNA transcribed from the inserted gene is translated into functional β -globin polypeptides.

- A** 1 only
- B** 2 only
- C** 3 only
- D** 1 and 2 only
- E** 1 and 3 only
- F** 2 and 3 only
- G** 1, 2 and 3

- 6 The diagram represents the secondary structure of a transfer RNA (tRNA) molecule.

Which row in the table correctly identifies parts 1, 2, and 3, and describes the overall molecular nature of tRNA?

	Part 1	Type of bond at 2	Part 3	Overall structure
A	amino acid attachment site	phosphodiester	anticodon	single-stranded RNA
B	amino acid attachment site	hydrogen	codon	single-stranded RNA
C	amino acid attachment site	hydrogen	anticodon	double-stranded RNA
D	amino acid attachment site	hydrogen	anticodon	single-stranded RNA
E	ribosome attachment site	phosphodiester	codon	double-stranded RNA
F	ribosome attachment site	hydrogen	anticodon	single-stranded RNA
G	mRNA attachment site	hydrogen	codon	single-stranded RNA
H	mRNA attachment site	phosphodiester	anticodon	double-stranded RNA

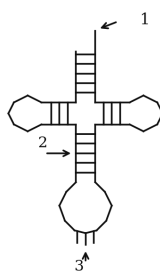


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- 7 The diagram shows the blood flow between four major mammalian organ systems, represented by boxes **W**, **X**, **Y**, and **Z**. Arrows indicate the direction of blood flow in the vessels connecting them.

Which row in the table correctly identifies boxes **W**, **X**, **Y**, and **Z**?

	Heart and lungs	Kidneys	Liver	Small intestine
A	X	Z	Y	W
B	Y	W	X	Z
C	Y	Z	W	X
D	Y	Z	X	W
E	Z	W	Y	X
F	Z	X	W	Y
G	X	W	Z	Y
H	Y	X	Z	W

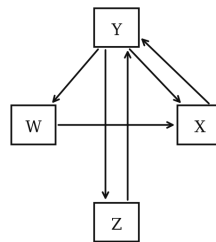


Diagram not to scale

- 8 An investigation was carried out using a single clone of the water flea *Daphnia magna* (all individuals were genetically identical).

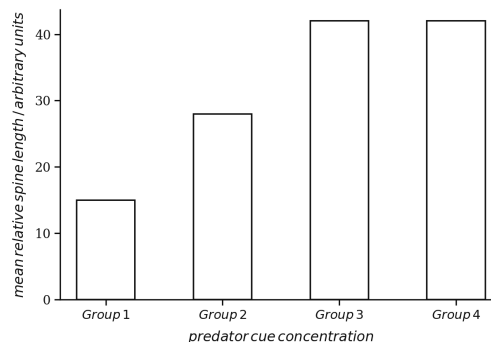
Forty newly hatched *Daphnia* of identical size were divided equally into four separate aquaria (Groups 1 to 4). The environmental conditions in the aquaria were controlled as follows:

- kept at the same constant water temperature
- supplied with the same daily quantity of algal food
- kept under the same light–dark cycle
- exposed to a different concentration of chemical cues (kairomones) released by predatory fish, where the concentration in Group 1 < Group 2 < Group 3 < Group 4

After reaching maturity, the relative length of the defensive tail spine was measured for each individual, and the mean relative spine length was calculated for each group. The results are shown in the graph below.

Which of the following statements could explain the results?

- 1 The difference in mean spine length between Group 1 and Group 2 is due to an environmental factor.
- 2 The mean spine length in Group 4 equals that in Group 3 because an environmental factor other than predator cue concentration is limiting growth.
- 3 The mean spine length in Group 4 equals that in Group 3 because the maximum spine length is constrained by the genotype of the clone.



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 9 Red-green colour blindness is a sex-linked (X-linked) recessive condition in humans.

A survey of a group of 1000 students recorded the following data:

Group	Phenotype / status	Number of individuals
Males	Colour-blind	48
Males	Normal vision	552
Females	Colour-blind	4
Females	Carrier with normal vision	72
Females	Non-carrier with normal vision	324

One nucleated white blood cell was collected from every individual in the group.

What is the total number of red-green colour blindness alleles present across all of these collected cells?

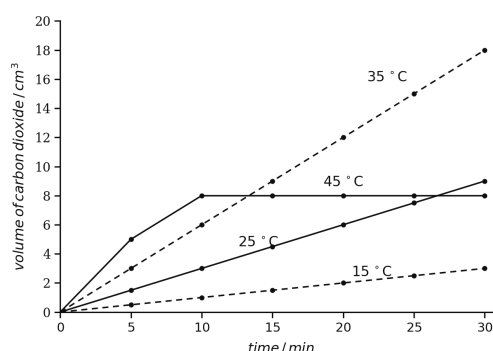
- A 56
- B 124
- C 128
- D 176
- E 200
- F 248

- 10** An investigation was carried out into the effect of temperature on anaerobic respiration in yeast. A suspension of yeast cells was supplied with excess glucose solution and incubated at four different temperatures: 15 °C, 25 °C, 35 °C, and 45 °C. In each case, the cumulative volume of carbon dioxide released was recorded at regular intervals over a period of 30 minutes.

All other variables were kept constant. The results are shown on the graph below.

Which of the following statements is/are correct?

- 1 During the first 10 minutes, the average rate of carbon dioxide production was greatest at 45 °C.
- 2 For the reaction at 35 °C, the average rate of carbon dioxide production was $0.01 \text{ cm}^3 \text{ s}^{-1}$.
- 3 Between $t = 15$ minutes and $t = 30$ minutes, the rate of carbon dioxide production at 45 °C was greater than at 15 °C.



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

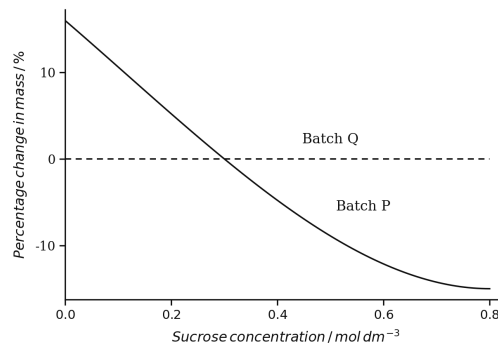
11 Identical cylinders of potato tissue were prepared and divided into two batches:

- Batch P consisted of untreated, living tissue.
- Batch Q was placed in boiling water for 10 minutes to kill the cells and then cooled to room temperature.

Both batches were immersed in sucrose solutions of concentrations ranging from 0.0 mol dm^{-3} to 0.8 mol dm^{-3} at 20°C for 2 hours. The percentage change in mass was measured, and the results are shown in the graph below.

Which of the following statements is/are correct?

- 1 At a sucrose concentration of 0.30 mol dm^{-3} , the water potential of the tissue in Batch P is equal to the water potential of the surrounding solution.
- 2 In Batch P, the pressure potential (Ψ_p) of the cells immersed in the 0.80 mol dm^{-3} sucrose solution is zero.
- 3 The lack of mass change in Batch Q is due to the breakdown of cellulose cell walls during boiling.



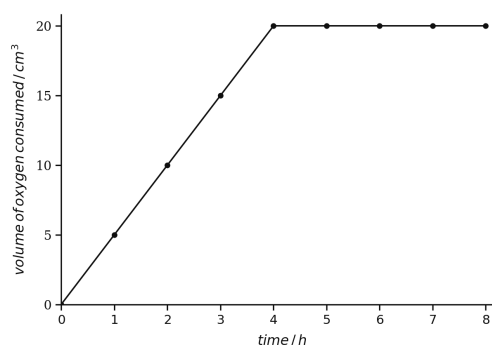
- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 12 A student investigated the rate of aerobic respiration in germinating pea seeds using a sealed respirometer containing potassium hydroxide solution to absorb CO_2 . The respirometer contained 50 g of germinating seeds and was maintained at a constant temperature of 20 °C.

The graph shows the cumulative volume of oxygen consumed over an 8-hour period.

Which row in the table gives the correct mean rate of oxygen consumption per gram of seeds between 0 and 4 hours, and the most likely reason why oxygen consumption stopped after 4 hours?

	Mean rate of oxygen consumption / $\frac{\text{cm}^3}{\text{g h}}$	Reason for oxygen consumption stopping after 4 h
A	0.050	available oxygen in the respirometer has been depleted
B	0.050	respiratory enzymes have denatured
C	0.10	available oxygen in the respirometer has been depleted
D	0.10	all respiratory enzyme active sites are saturated
E	0.40	available oxygen in the respirometer has been depleted
F	0.40	respiratory enzymes have denatured
G	5.0	available oxygen in the respirometer has been depleted
H	5.0	all respiratory enzyme active sites are saturated



- 13** Two stem cuttings taken from the same parent rose plant are grown in two different gardens. Which of the following could lead to phenotypic variation between the two mature plants?
- 1 differences in the intensity of light received by the plants
 - 2 crossing over during meiosis in the parent plant
 - 3 spontaneous mutations occurring during mitotic cell divisions
- A** none of them
 - B** 1 only
 - C** 2 only
 - D** 3 only
 - E** 1 and 2 only
 - F** 1 and 3 only
 - G** 2 and 3 only
 - H** 1, 2 and 3

- 14** An ecologist investigated the population size of an insect species in an isolated woodland using the mark-release-recapture method.

On Day 0, 200 insects were captured, marked on the thorax with waterproof paint, and released back into the woodland.

On each of the following five days (Days 1 to 5), a random sample of 80 insects was captured from the woodland. The number of marked individuals (R) in each sample was recorded, and all 80 insects were immediately released back into the habitat.

The graph shows the number of marked insects recaptured on each day.

Population size (N) can be estimated using the Lincoln index:

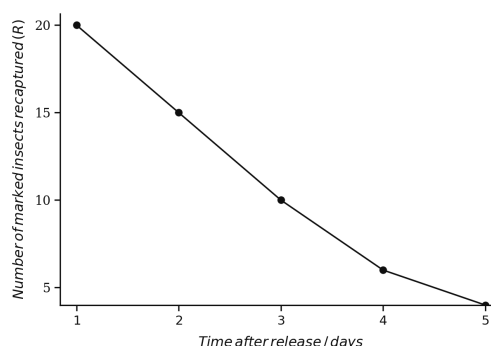
$$N = \frac{M \times C}{R}$$

where:

- M is the number of individuals initially marked on Day 0
- C is the total number of individuals captured in the recapture sample
- R is the number of marked individuals recaptured

Which of the following statements is/are correct?

- 1 Based on the sample collected on Day 1, the estimated population size in the woodland is 800.
- 2 If the paint made marked insects more conspicuous to predators, the Lincoln index would progressively overestimate the true population size over the five days.
- 3 If the insects became increasingly 'trap-shy' after their initial capture, the Lincoln index calculated on subsequent days would underestimate the true population size.



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 15** The diagram shows a cross-section through part of a dicotyledonous leaf, showing different cellular layers.

Which of the following statements is/are correct?

- 1 The leaf is an organ, and Layer 1 and Layer 2 are each tissues.
- 2 The nucleus of a living cell in Layer 1 contains the genes that code for the enzymes required for chlorophyll synthesis.
- 3 When illuminated with white light, photophosphorylation occurs in the cells of both Layer 1 and Layer 2.

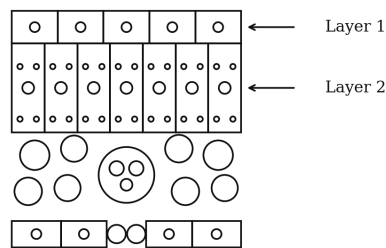


Diagram not to scale

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

16 The following statements describe various interspecific ecological interactions:

1 *Rhizobium* bacteria living inside root nodules of a legume, obtaining carbohydrates while supplying the plant with fixed nitrogen compounds

2 an epiphytic orchid growing on the high branch of a rainforest tree to access light, without absorbing nutrients or water from the tree

3 cleaner wrasse removing and feeding on ectoparasites from the skin and gills of a coral trout

4 mistletoe inserting specialized structures into the vascular tissues of a host tree to extract water and mineral ions

5 photosynthetic dinoflagellates (zooxanthellae) living inside reef-building coral cells, providing organic nutrients in exchange for inorganic compounds and shelter

Which of the statements describe(s) an example of mutualism?

A 1 only

B 3 only

C 1 and 5 only

D 2 and 4 only

E 3 and 5 only

F 1, 2 and 3 only

G 1, 3 and 5 only

H 2, 4 and 5 only

- 17 The enzyme β -galactosidase catalyses the hydrolysis of ortho-nitrophenyl- β -D-galactopyranoside (ONPG) into galactose and ortho-nitrophenol (ONP).

* ONPG is colourless in solution.

* ONP is colourless at neutral and acidic pH, but turns bright yellow under alkaline conditions (absorbance measured at 420 nm).

Five test tubes were prepared containing equal volumes and concentrations of ONPG and β -galactosidase at pH 7.0. Each tube was incubated at a different temperature for 10 minutes.

After 10 minutes, an equal volume of concentrated Na_2CO_3 solution was added to each tube to raise the pH to 10.5. The absorbance of each solution at 420 nm was measured immediately and then measured again after 10 minutes at room temperature.

Tube	Incubation temperature / $^{\circ}\text{C}$	Absorbance at 420 nm immediately after Na_2CO_3 addition	Absorbance at 420 nm after 10 minutes
1	10	0.14	0.14
2	25	0.41	0.41
3	40	0.86	0.86
4	55	0.35	0.35
5	70	0.03	0.03

Which of the following statements is/are correct?

1 The addition of Na_2CO_3 solution halted the catalytic activity of the β -galactosidase.

2 At the end of the initial 10-minute incubation, the concentration of unreacted substrate remaining was lowest in tube 3.

3 The optimum temperature for this β -galactosidase must be 40°C .

- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

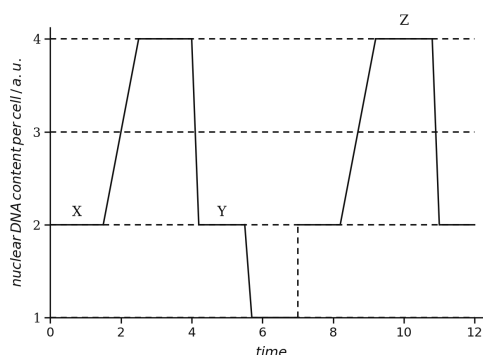
- 18** The graph shows changes in the nuclear DNA content of a single cell lineage during spermatogenesis in a human male, followed by fertilisation and the first mitotic division of the resulting zygote.

The male is heterozygous (Aa) for an autosomal gene. During meiosis I, the chromosome carrying allele A segregates into the cell lineage shown. The resulting mature sperm cell fertilises an ovum from a female who is homozygous (AA) for this gene.

Assume that no crossing over or mutations occur.

How many copies of allele A are present in the single cell at each of the stages marked **X**, **Y**, and **Z** on the graph?

	stage X	stage Y	stage Z
A	1	1	2
B	1	1	4
C	1	2	2
D	1	2	4
E	2	1	2
F	2	1	4
G	2	2	2
H	2	2	4



- 19** A culture of yeast cells is suspended in an aqueous nutrient medium.

The table below shows the concentrations of two dissolved solutes, substance **P** and substance **Q**, as well as the total solute concentration, in the nutrient medium and in the cytoplasm of the yeast cells.

	Concentration in medium / mmol dm ⁻³	Concentration in cytoplasm / mmol dm ⁻³
substance P	0.2	5.0
substance Q	15.0	1.5
total solutes	30.0	180.0

Under normal aerobic conditions, there is a net movement of substance **P**, substance **Q**, and water molecules into the yeast cells.

The culture is then treated with a metabolic poison that completely prevents the synthesis of ATP.

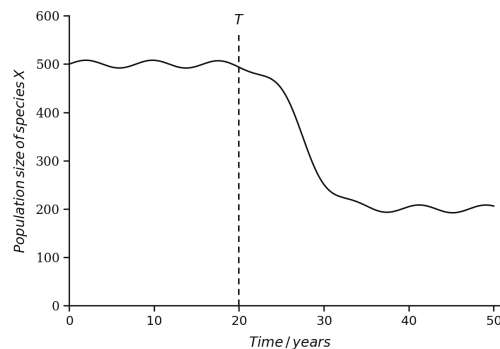
Which row in the table below correctly shows the net movement of substance **P**, substance **Q**, and water molecules across the cell membrane immediately after the addition of this poison?

	Net movement of substance P	Net movement of substance Q	Net movement of water
A	moves into the cell	moves into the cell	moves into the cell
B	moves into the cell	moves into the cell	moves out of the cell
C	moves into the cell	no net movement into the cell	moves into the cell
D	moves into the cell	no net movement into the cell	moves out of the cell
E	no net movement into the cell	moves into the cell	moves into the cell
F	no net movement into the cell	moves into the cell	moves out of the cell
G	no net movement into the cell	no net movement into the cell	moves into the cell
H	no net movement into the cell	no net movement into the cell	moves out of the cell

- 20** The graph shows changes in the population size of a small rodent species (species X) on an isolated island over a 50-year period. At time T , a non-native mammal species (species Y) was introduced to the island.

Which of the following could account for the change in the population of species X observed after time T ?

- 1 Species Y competed with species X for some, but not all, of its food resources.
- 2 Species Y was competitively superior and occupied the exact same ecological niche as species X.
- 3 Species Y preyed on species X, but species X had access to a limited number of safe refuges.



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 21** An intact, living plant cell is surrounded by an aqueous solution.

Which of the following will **always** result in a net movement of water into the cell by osmosis?

1 transferring the cell into a solution that has a higher water potential than the solute potential of the cell's cytoplasm

2 increasing the concentration of dissolved solutes inside the cell's central vacuole

3 increasing the water potential of the surrounding solution to a value greater than the total water potential of the cell

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 22** During hot, dry conditions, a terrestrial plant closes its stomata while remaining in bright daylight.

Which of the following could be a direct or indirect consequence of this stomatal closure?

1 A decrease in the rate of transpiration from the leaves.

2 A reduction in the concentration gradient of carbon dioxide between the surrounding air and the leaf air spaces.

3 An increase in the temperature of the leaf tissue.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 23** An intracellular enzyme, enzyme X, is present in the cytoplasm of both mammalian cells and *Escherichia coli* (*E. coli*) bacteria. Enzyme X catalyses the conversion of a colourless substrate, S, into a fluorescent product, P.

Neither enzyme X nor substrate S can cross an intact cell surface membrane.

Lysozyme is an enzyme that digests the peptidoglycan cell wall of bacteria.

Four separate suspensions were prepared:

1. Mammalian cells suspended in pure water
2. Intact *E. coli* bacteria suspended in pure water
3. *E. coli* bacteria pre-treated with lysozyme to remove their cell walls, then suspended in pure water
4. *E. coli* bacteria pre-treated with lysozyme to remove their cell walls, then suspended in an isotonic buffer solution

Substrate S was added to each suspension.

Which row in the table correctly shows whether fluorescent product P will be detected in each mixture?

Key:

✓ = product P detected

X = product P not detected

	Mammalian cells in pure water	Intact *E. coli* in pure water	Lysozyme-treated *E. coli* in pure water	Lysozyme-treated *E. coli* in isotonic buffer
A	✓	X	✓	X
B	✓	✓	✓	X
C	✓	X	X	X
D	✓	X	✓	✓
E	X	X	✓	X
F	X	✓	X	✓
G	✓	✓	X	X
H	X	X	X	X

- 24** The epithelial lining of the mammalian small intestine undergoes continuous renewal. Adult stem cells located at the base of intestinal crypts divide, and their daughter cells differentiate as they migrate along the villi.

A healthy adult has an estimated total intestinal epithelial surface area of 0.30 m^2 .

A biopsy shows that there are an average of 4.0×10^5 epithelial cells per mm^2 of this surface.

The total number of epithelial cells is maintained at a constant level, and the average lifespan of an epithelial cell is 5.0 days (120 hours).

Which of the following statements is/are correct?

- 1 The average rate of new epithelial cell production is 1.0×10^9 cells per hour.
- 2 The stem cells at the base of the crypts divide by meiosis to maintain the cell population.
- 3 Mitochondria in mature intestinal epithelial cells produce the ATP required for the active transport of nutrients.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 25** Three stages of aerobic cellular respiration in human cells, X, Y and Z, are compared in the table below:

stage of respiration	releases CO_2	directly consumes molecular O_2	produces reduced NAD (NADH)	synthesises ATP
X	no	no	yes	yes
Y	yes	no	yes	yes
Z	no	yes	no	yes

Which row in the table below correctly identifies stages X, Y and Z?

	X	Y	Z
A	glycolysis	Krebs cycle	oxidative phosphorylation
B	glycolysis	oxidative phosphorylation	Krebs cycle
C	Krebs cycle	glycolysis	oxidative phosphorylation
D	Krebs cycle	oxidative phosphorylation	glycolysis
E	oxidative phosphorylation	glycolysis	Krebs cycle
F	oxidative phosphorylation	Krebs cycle	glycolysis

- 26** Consider the following three cell types:

- a bacterium (*Escherichia coli*)
- a unicellular fungus (yeast, *Saccharomyces cerevisiae*)
- a plant palisade mesophyll cell

Which of the following features is/are present in **all three** of these cell types?

- 1 circular DNA
- 2 70S ribosomes
- 3 membrane-bound organelles
- 4 enzyme-catalysed synthesis of ATP

- A** 4 only
- B** 1 and 2 only
- C** 1 and 4 only
- D** 2 and 4 only
- E** 3 and 4 only
- F** 1, 2 and 4 only
- G** 2, 3 and 4 only
- H** 1, 2, 3 and 4

27 Which of the following mature cell types contain a nucleus?

- 1 mature companion cells in phloem
- 2 mature human red blood cells
- 3 mature phloem sieve tube elements
- 4 mature leaf guard cells

- A** 1 and 2 only
- B** 1 and 4 only
- C** 2 and 3 only
- D** 2 and 4 only
- E** 3 and 4 only
- F** 1, 2 and 3 only
- G** 1, 3 and 4 only
- H** 1, 2, 3 and 4